

Instagram Search Capabilities

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Abstract—Instagram is a social media platform that has been widely used over the past decade. Users typically use the app as a photo sharing tool for social networking. Within Instagram, the app has a dedicated page known as the “Explore” page that is used to display personalized recommendations of posts on the app for the user to see. This page is specifically designed using Instagram’s personal recommendation systems.

1 PROBLEM SPACE

1.1 Problem and Context

Instagram’s constantly expanding user base has been a great contribution to how Instagram has been able to solidify its position as one of the most popular social media platforms today. With more users utilizing the app, Instagram must ensure that every aspect of its platform is able to give the user the best experience. The app’s Explore page is one of those aspects of the app that I believe can still be improved. Currently, the explore page is a culmination of an infinite scrolling grid-like design of posts within the app that is generated by Instagram’s recommendation system to show the user.

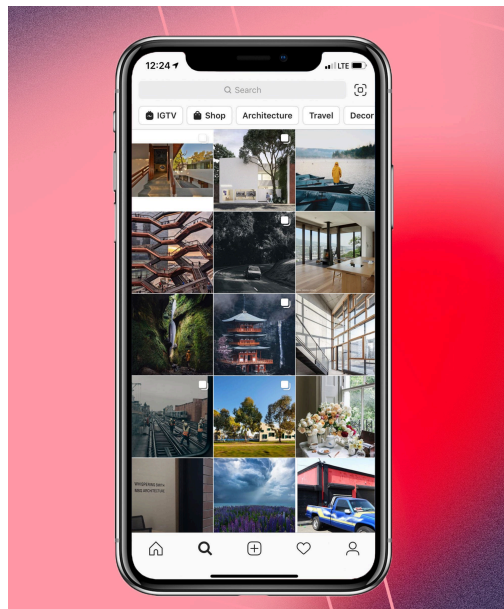


Figure 1—Image of Instagram’s Explore page. [Source](#)

The explore page is developed and filled with posts using the recommendation system that utilizes advanced machine learning models to generate personalized recommendations of posts to the user (Scaling the Instagram Explore Recommendations System, 2023). Although the system Instagram is currently using is very complex, there are still many instances where it can be improved. Most noticeably is the content diversity and personalization accuracy of the posts that are displayed within the explore page. The content diversity of the page revolves around the user's experience with the different types of posts while using the explore page. The lack of differences in the types of content that may come up may cause the user's interests to wane and cause the user to lose focus on the materials he or she is engaging in. In addition, the personalization accuracy stems from the lack of control the user has over how the explore page is displayed. If the user is not able to have at least some type of control over the personalization of the page or how the recommendations are generated there is a high chance that posts and topics that do not interest the user will be placed within the explore page as a recommendation. Although Instagram's current explore page is very well done, there are still many changes that can enhance the user's overall experience with the app.

1.2 Proposed Design

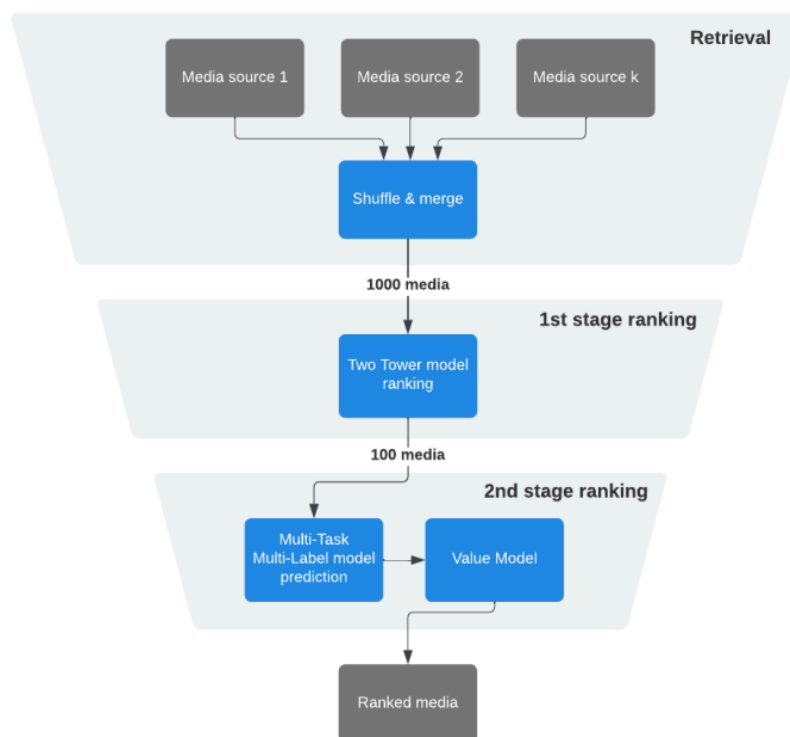


Figure 2—Instagram's ranking system for explore page media. [Source](#)

Instagram’s current recommendations system utilizes a ranking system for all the media within the app to be able to accumulate an appropriate group of recommendations to offer to the user. The higher ranked media files are identified from the users which will then be used to develop the two tower neural networks, which will then be used to compare between different media for the overall ranked media.

My proposed solution encompasses giving the user control over one of the steps of the ranking system. Since the machine learning model is taking specific aspects from the user that can be thought of as high ranking to develop the ranked media, if the user is given control over what aspects are taken into account and what posts are used to develop the model then the user is able to have a higher level of personalization towards the recommendations in the explore page. In addition, since the user is able to adjust the content at will, the explore page will be able to have added content diversity since the user will be able to change the posts themselves.

1.3 Comparison

This redesign is targeting Instagram's explore page’s current lack of content diversity and the lack of personalization accuracy. By allowing users to have at least some control over how the modeling is trained for the recommendation system, the results will be able to have increased personalization accuracy since the users themselves are controlling what they want to be shown and the content will be able to be diversified based on what the user has chosen. In addition, the user can control the recommendation system to offer posts that he or she does not usually view for more content diversity.

2 NEED FINDING EXERCISE

2.1 Need Finding

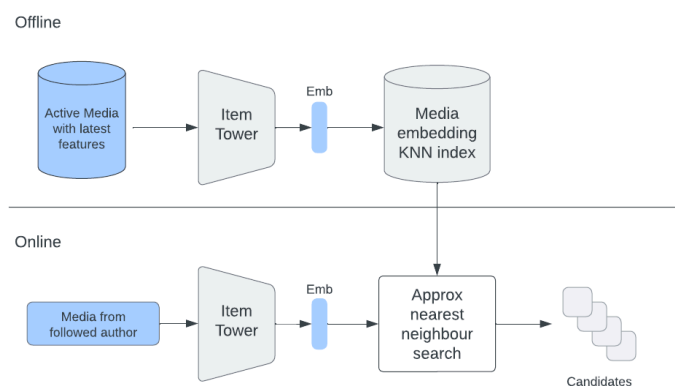


Figure 3—User Interaction History for model. [Source](#)

For this design plan, the recommendation system will have to be altered by the user. The current model takes into account what posts the user has seen, interacted with, liked, saved, etc to train the model. To account for the ability for the user to be able to alter what is taken into account for the features and media that is used in the neural network, an entire new page needs to be created for the personalization.

2.2 Functional Components

The new personalization page that is added within the explore page will give adequate control of the media taken into account for the recommendation. This means that the user will have to be able to individually mark posts that are included and excluded from the model. These pages will have a very similar setup to how the interactions page in Instagram's setting is set up; however, these pages will be specifically for exploring page customization. Once this process is done then the entire rest of the process will be exactly the same as the current explore page recommendation system.

2.3 Architecture

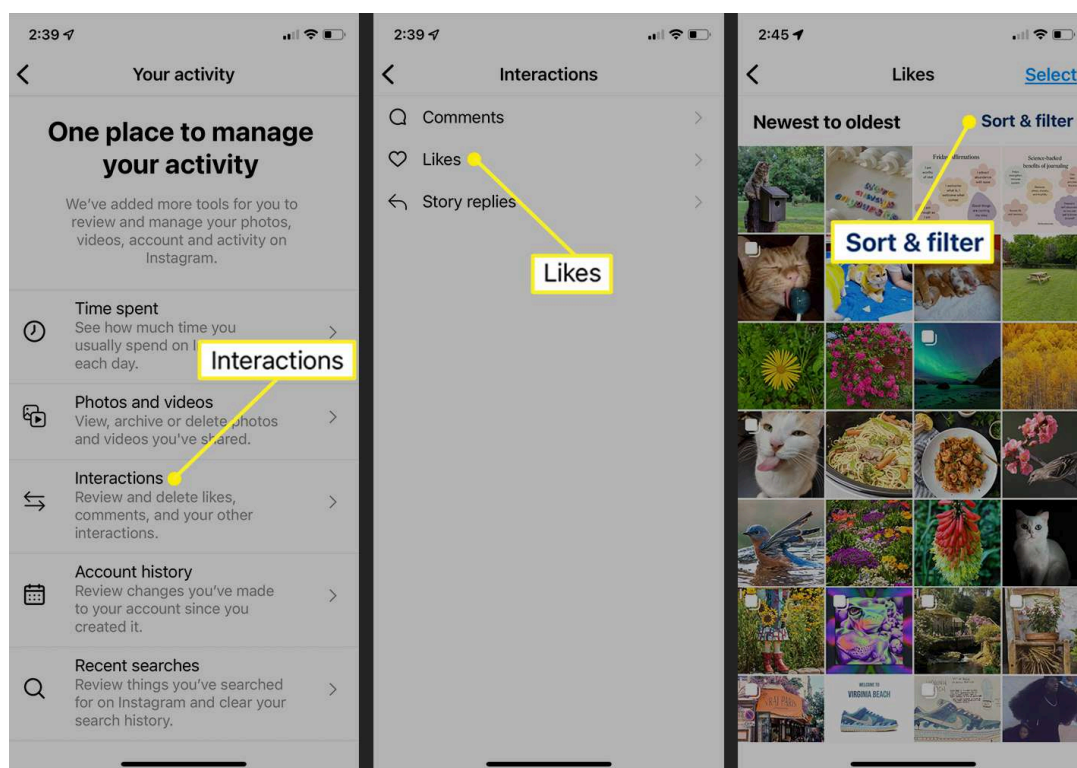


Figure 4—Instagram's Interactions page to Likes page. [Source](#)

The explore customization page will follow this similar setup, with the addition of the user's ability to pick and choose posts for the model training.

2.4 Data Collection

The data collected from this redesign is heavily qualitative. The data that is being collected is the accuracy in which the personalization of the explore page is accurate of which the user is trying to render. The main data that needs to be collected is whether or not the first 100 posts of the returned recommendation system of the new model is accurate of the different topics and media that the user has chosen.

2.5 Utility Types

This redesign is focused on improving the overall functionality of the explore page. Although the previous explore page should be able to express some personalization of the posts for the user, the ability for the user to control the customization of the posts themselves gives the page the most possible personalization of the user, and increases the accuracy of said personalization compared to the original.

3 DESIGN REFINEMENTS

3.1 Need Finding

From a cursory glance at Instagram's explore page, there seems to be a good amount of content diversity throughout the page. However, this is only truly accurate with new users since the recommendation system will immediately start collecting data to train the system to offer the user more personalized content. This takes the cold start issue into account, since with a cold start there is not enough sufficient data or interaction history for the recommendation system to offer relevant posts to the user. Since this is the case with a cold start, the recommendation system would offer a completely diverse selection of posts entirely based on current popular posts instead. Although this may be counterintuitive to a recommendation system, the bias that may arise from the recommendations can completely skew the different types of posts a user sees on the explore page towards certain topics. If a user wants to see other types of content the user would have to search for it specifically.

3.2 Functional Components

To counteract the issue with the lack of content diversity, the redesign involves including a simple option for the users to simply turn off the recommendation system on the explore page and see the explore page entirely as a new user instead. This will ultimately give the user the option to be able to not be affected by the recommendation system that is already set in place and be able to see posts and types of content that the user would normally not be exposed to.

3.3 Architecture

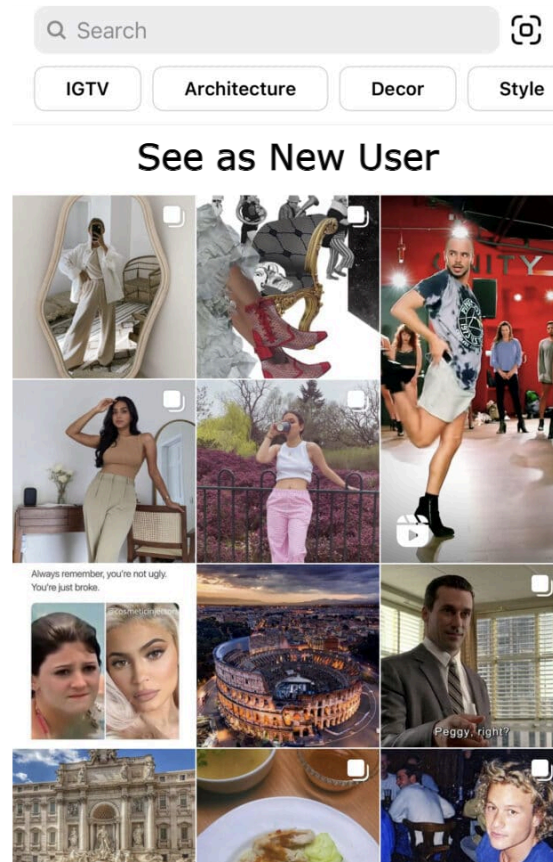


Figure 5—Instagram’s Explore Redesign

3.4 Data Collection

Similarly to the previous redesign, this redesign’s data collection will also be entirely qualitative. The only data collection will be entirely through user satisfaction. Since there is no quantitative aspect of the redesign that can be analyzed, the determination in which the redesign is successful will be entirely based on user feedback and reviews.

3.5 Utility Types

The purpose of this redesign is to alleviate the issue of the lack of content diversity within the explore page of Instagram. With the inclusion of both redesigns, the explore page is able to have more flexibility in the customization of the posts that are recommended to the user and the user is able to remove any bias that the system may have towards the user in its recommendation. The lack of content diversity can be easily resolved since the user now has many options to be able to see new and different content as he or she wishes.

4 ADDITIONAL DESIGN CONSIDERATIONS

4.1 Evaluation of Baseline

The baseline designs are evaluated on the functionality and flexibility of the designs to handle the two major issues that were mentioned throughout this paper. The performance of both redesigns are heavily evaluated based on qualitative data gathered solely through user feedback. Since neither of the redesigns have any quantitative aspects for evaluation, the success of the designs are entirely dependent on how users feel about the designs. This data would mainly be collected through surveys and reports to determine if the personalization accuracy and lack of content diversity issues have been resolved.

4.2 Additional Requirements

For the baseline designs, additional requirements for the designs would be to increase the simplicity of the designs while ensuring the functionality and error handling of the design. Since the first redesign requires the user to have to manually change any posts or features that are included in the training of the model, this would leave the users with possibly hundreds and thousands of posts to go through. The design would need to be able to give the user easier control and simplicity in how the design worked to mark each post so that the user would not have to mark each individual post. This would ultimately lead to possible further redesign of how the page is laid out for the user to see.

5 REFERENCES

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